

Deva Anand

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Education

University of Massachusetts Amherst

Master of Science in Computer Science

September 2025 – May 2027 (Expected)

Amherst, MA

Coursework: Advanced Machine Learning, Performance Engineering, Optimization in CS, Systems for Data Science, Advanced NLP

Vels Institute of Science, Technology and Advanced Studies

Bachelor of Engineering in Computer Science and Engineering

August 2019 – May 2023

Chennai, India

Skills

Languages

Python, C++, Java, JavaScript, TypeScript, SQL, Bash

ML & Vision

PyTorch, JAX, NumPy, scikit-learn, CNNs / ResNet, transfer learning, MediaPipe, training pipelines, model optimization, autodiff

Systems & Edge

On-device / edge inference, async pipelines & queues, whisper.cpp, llama.cpp, Linux profiling, performance optimization

Testing & Tooling

pytest, integration testing, automated test suites, Git, GitHub Actions, CI/CD, Docker

Backend & Cloud

FastAPI, Node.js, Express, REST APIs, AWS, GCP, Azure

Experience

Wysa

Backend Engineer

July 2023 – July 2025

Bengaluru, India

- Engineered a multi-tenant production backend (Node.js, MongoDB) for **8+ UK clients** processing **10,000+ monthly submissions**; owned reliability across releases, debugged and resolved tenant-specific production incidents, and remediated **20+ VAPT findings**.
- Architected a full-stack **ML training-data annotation platform** (React, Node.js, MongoDB) used daily by the AI team to label data for production models; cut manual annotation work by **100+ hours/month** and improved labeling throughput by **60%**.

Cario Growth Services Pvt. Ltd.

Backend Developer Intern

November 2022 – May 2023

Chennai, India

- Designed **20+ RESTful APIs** in Node.js + Fastify over PostgreSQL and integrated LLM-powered context-aware replies into a chat product, iterating on prompt design and response-quality heuristics for a production customer-facing experience.
- Migrated the core codebase from **JavaScript to TypeScript** across multiple repositories, improving type safety and long-term maintainability through targeted refactoring.

Projects

Sonare - Real-Time On-Device Perception App

React, Electron, FastAPI, MediaPipe, whisper.cpp | Qualcomm Edge AI Hackathon 2025

github.com/Deva-1903/Qualcomm-Sep25-Team-Sonare

- Built a cross-platform app with an **on-device perception pipeline**, **MediaPipe hand tracking and gesture classification** from a live camera feed, **whisper.cpp** speech recognition, and local TTS, integrating camera, microphone, UI, and backend under hackathon time pressure.
- Engineered a FastAPI backend and Electron/React frontend running **fully offline on-device**; designed **async APIs and queues** to keep the UI responsive while processing live video and audio streams under tight device constraints.
- Added **temporal stabilization** to gesture classification, smoothing noisy per-frame predictions into consistent gestures across the live camera stream for a reliable on-device experience.

Deep Learning & Generative Models - From-Scratch ML

JAX, PyTorch, NumPy | UMass CS 689 Advanced Machine Learning

github.com/Deva-1903/UMASS_CICS_689

- Trained and benchmarked **5 neural architectures** (including a VGG-style CNN and ResNet) on **CIFAR-10** across **3 optimizers** with learning-rate tuning, documenting train/test-error curves.
- Built a **reverse-mode automatic differentiation engine** from scratch in NumPy validated against JAX to machine precision, and implemented a normalizing flow and DDPM diffusion model from scratch in JAX.
- Derived convergence behavior of **gradient descent and SGD** on PSD/PD objectives and verified estimator asymptotics empirically, strengthening the optimization foundations behind model training.

AgenticSearch - Tested, Provenance-First Pipeline

Python, FastAPI, OpenAI / Groq, SQLite

github.com/Deva-1903/ciir_agentic_search

- Built a multi-stage software pipeline (typed retrieval planning, deterministic extraction with LLM fallback, output verification) and **hardened it with 150+ automated tests**, provider fallback, and pipeline instrumentation for validation.
- Returned **cell-level provenance** for every result behind a reviewer-facing trust UI, and deployed the service as a live demo on **Google Cloud Run**.

Publication & Extracurriculars

- Publication:** “Alzheimer’s Disease Classification using Transfer Learning,” **IEEE CONIT 2023**, applied deep transfer learning (computer vision) on neuroimaging data for medical image classification.
- Selected for **GirlScript Summer of Code (GSSoC) 2023** and contributed to multiple open-source projects (Linkfree, Freehit, ProjectsHut).
- Tutored **30+** underprivileged students in programming at Sayur, non-profit in South Tamil Nadu.